

Pipeline Update Report

Round 3

Nanotech Media Analysis Project

April 30, 2026

12,206 articles | 7 sources | 1983-2005

Overview

This round imported existing Claude analysis from a prior integration file, normalized the qualitative coding columns for graphing, and built 13 new time-series charts showing how sources/communities differ across five qualitative dimensions: attitude toward nanotechnology, analogy presence, analogy temporality, sustaining vs disrupting framing, and funding arguments.

What was done:

- 1. Imported Claude analysis from old integration file: 10 columns, ~4,341 rows matched by Title + Sources key.
- 2. Normalized 5 graphing columns: Attitude, Analogy Present, Analogy Temporality (Coded), Sustaining vs Disrupting, Funding Argument Present.
- 3. Added 13 qualitative dimension time-series charts (3yr moving average, all sources overlaid).
- 4. Replaced per-source summary charts with combined all-sources overlay charts. Summary PDF now has 21 figures.

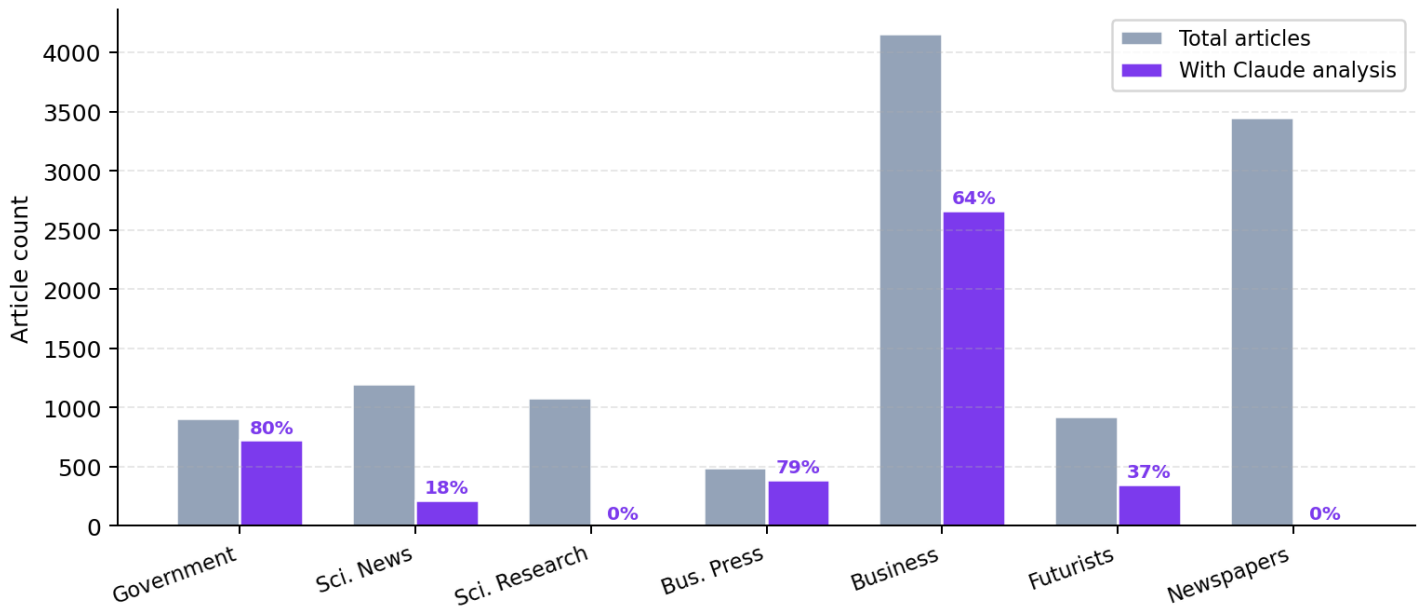
1. Imported Claude analysis from old file

The file "Old Python Files/Claude integration of output file.xlsx" contained prior Claude analysis for ~5,000 articles across its "Big dataset" sheet (10,156 rows, 42 columns). import_claude_columns.py matched rows by Title + Sources key and imported 10 Claude analysis columns into the current output.xlsx.

Imported columns:

- Sentiment Toward Nanotechnology (4,341 rows matched)
- Analogies, Purpose of Analogies (997 rows with text)
- Temporality of Analogies (162 unique free-text values)
- Analogy: Status Quo (Reinforcing/Disrupting/Both)
- Analogy: Funding Argument (Yes/No)
- Source Analyzed, Perspective on Technology, Mentions Nanotechnology, Actors Mentioned

Claude Analysis Coverage by Source (imported from old file)



Science Research and Newspapers have 0% coverage because they were not present in the old integration file. Science News has low coverage (18%) because most of its articles in the old file did not receive Claude analysis.

2. Normalized qualitative coding columns

The old file used different column names and value formats than the current coding schema. `import_claude_columns.py` normalizes these into 5 graph-ready columns:

New Column	Source	Mapping
Attitude	Sentiment Toward Nanotech	lowercase (Positive -> positive)
Analogy Present	Analogies	1 if text exists, 0 if null but coded
Analogy Temporality (Coded)	Temporality of Analogies	Free text -> past/present/future/mixed
Sustaining vs Disrupting	Analogy: Status Quo	Reinforcing -> sustaining
Funding Argument Present	Analogy: Funding Argument	Yes -> 1, No -> 0

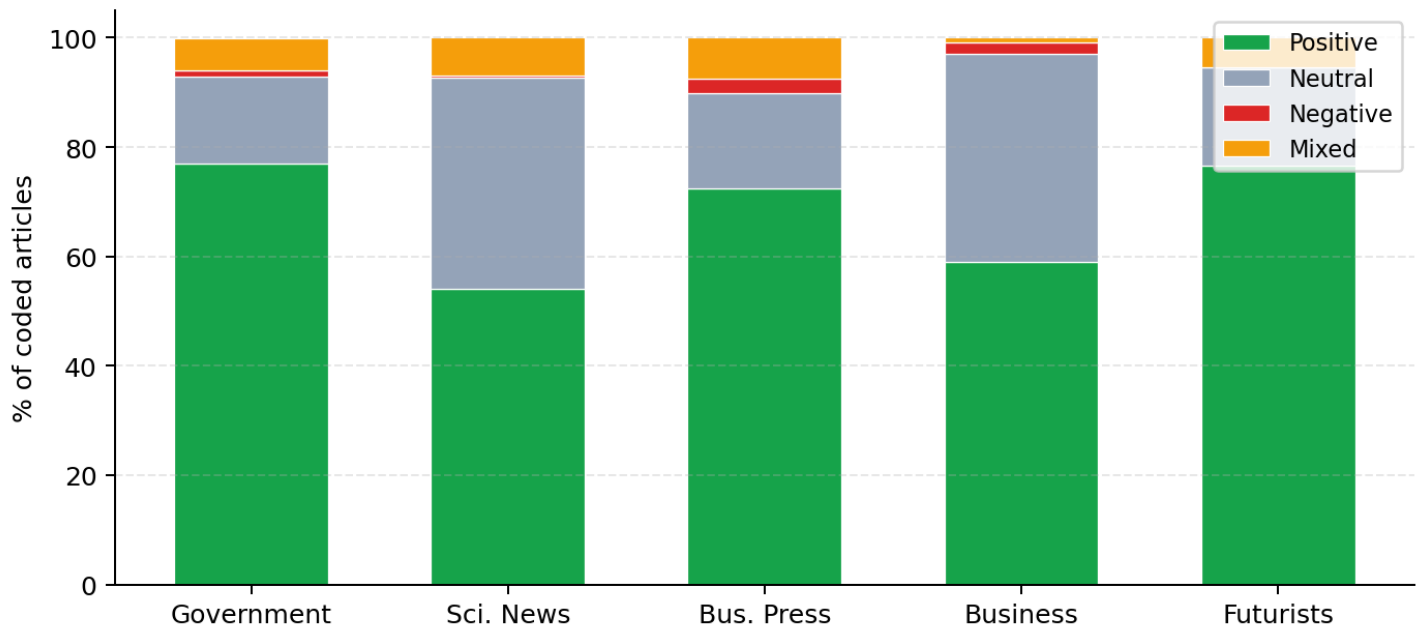
Value distributions after normalization:

Attitude: positive=2,795 neutral=1,340 negative=74 mixed=131
 Analogy: present=996 absent=3,344
 Status Quo: sustaining=1,923 disrupting=1,113 both=49
 Funding: yes=896 no=2,625

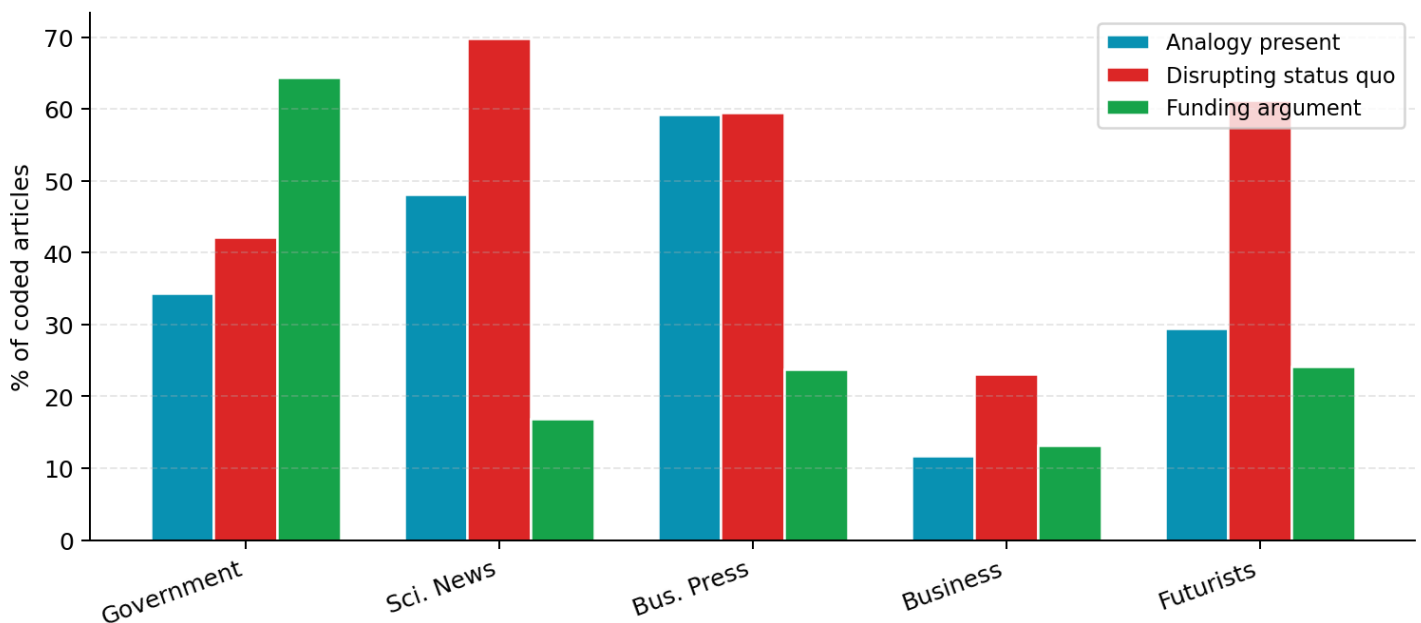
3. Qualitative dimension graphs over time

13 new time-series charts showing how sources/communities differ across 5 qualitative dimensions. Each chart overlays all sources with available data, using 3-year centered moving averages. Only sources with coded articles appear (Science Research and Newspapers are absent).

Attitude Toward Nanotechnology by Source

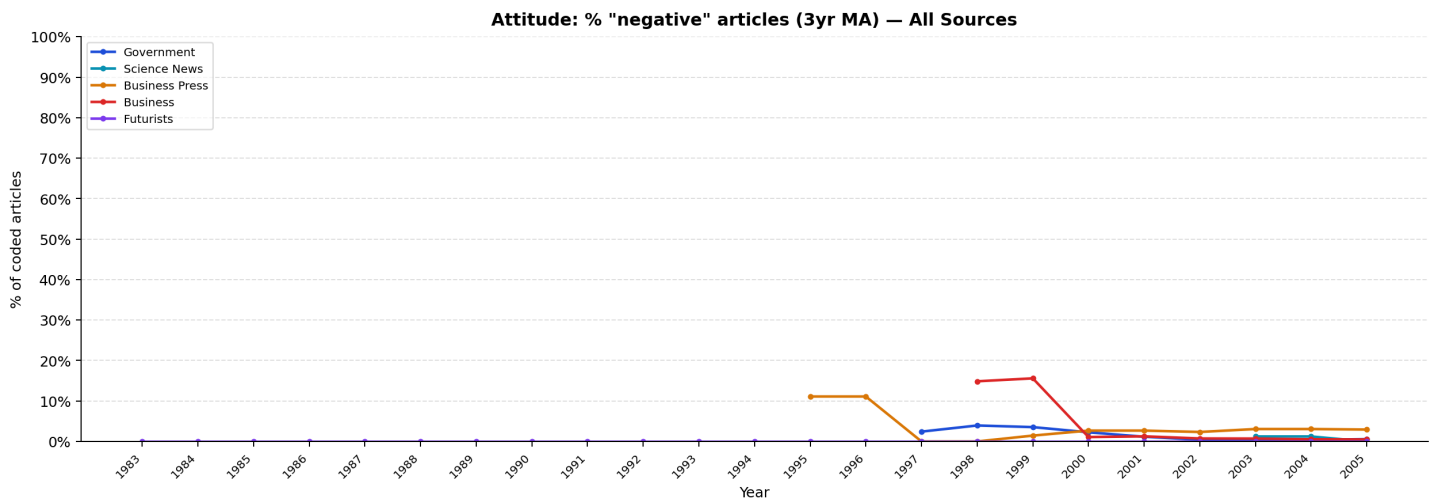
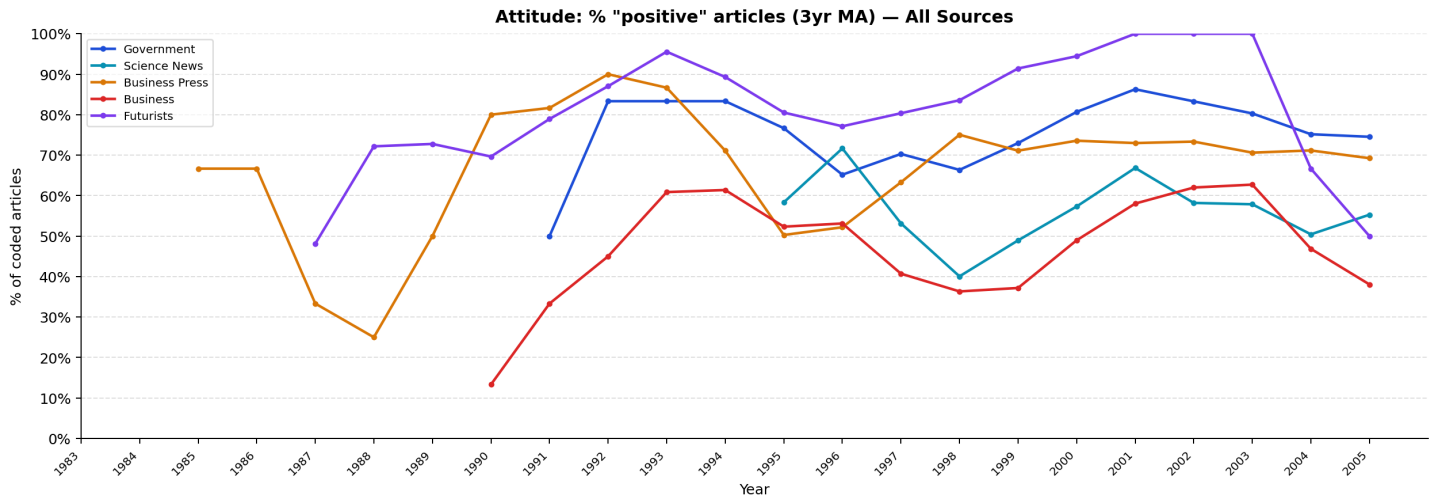


Qualitative Dimensions by Source



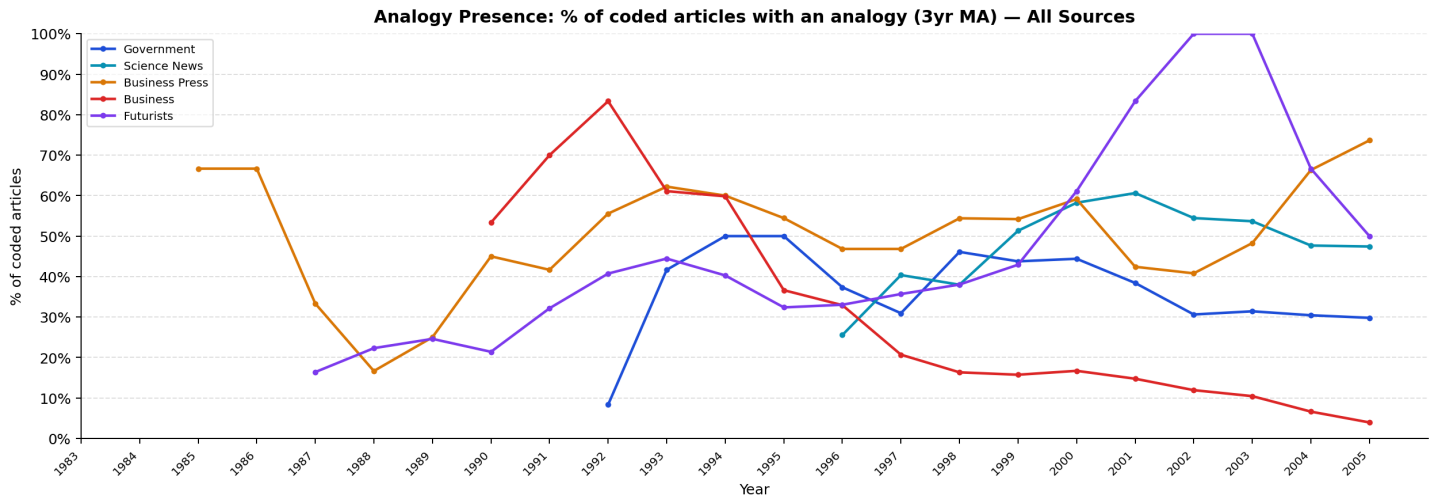
Attitude over time

Percentage of coded articles with positive vs negative attitude toward nanotechnology, per source, over time.



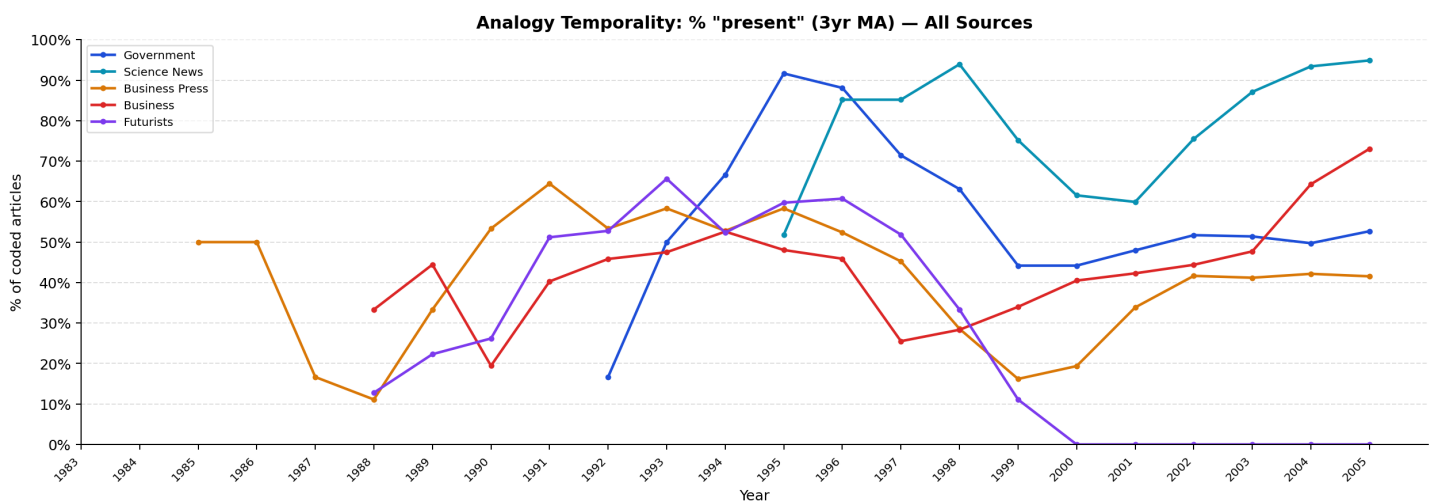
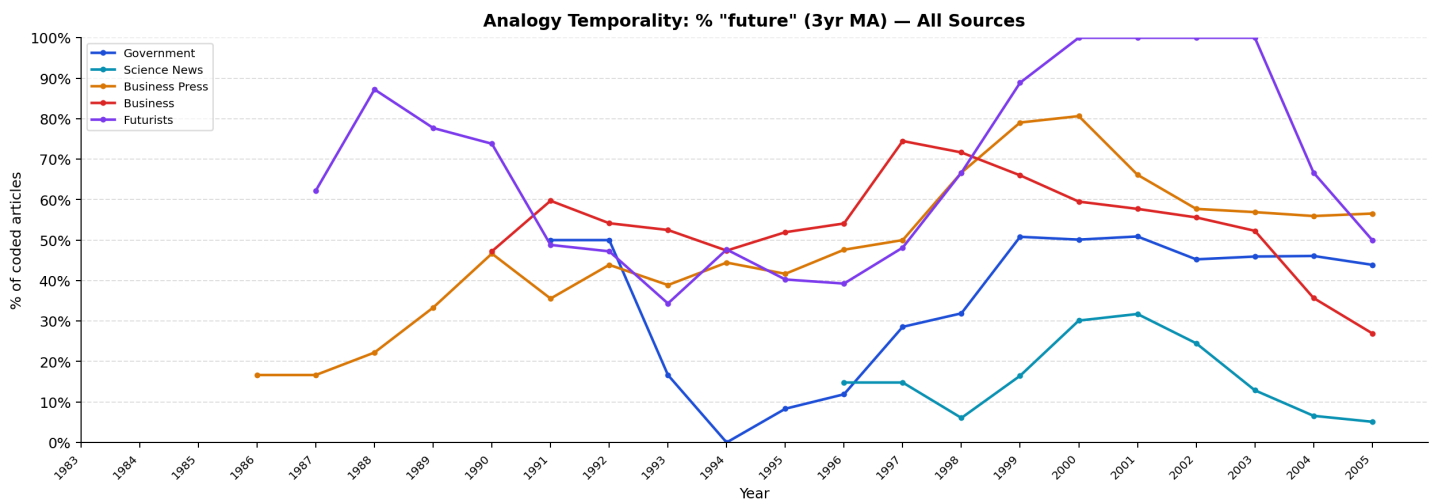
Analogy presence over time

Percentage of coded articles that use an analogy or metaphor to explain/frame nanotechnology.



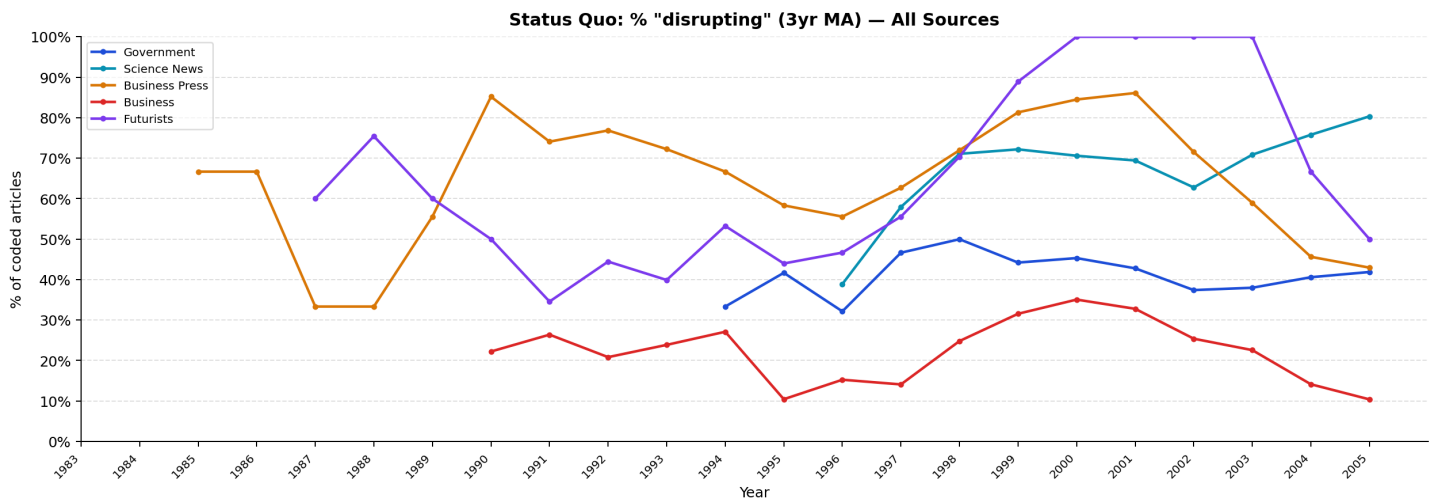
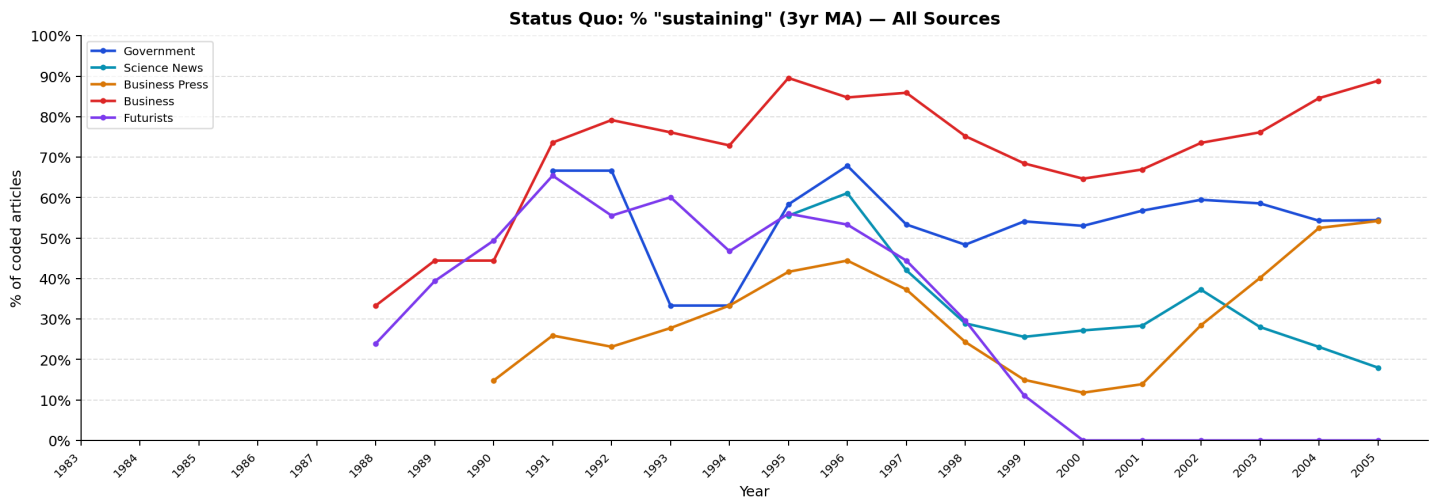
Analogy temporality over time

Among articles with analogies, the temporal orientation of the comparison.



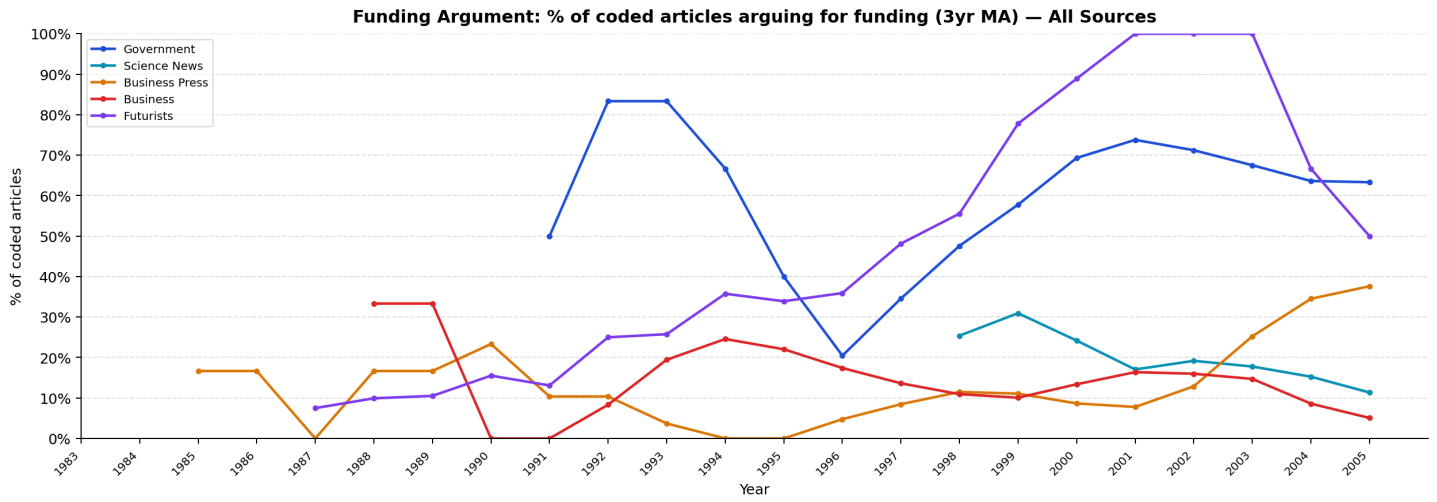
Sustaining vs Disrupting over time

Whether nanotechnology is framed as sustaining existing industries or disrupting the status quo.



Funding argument over time

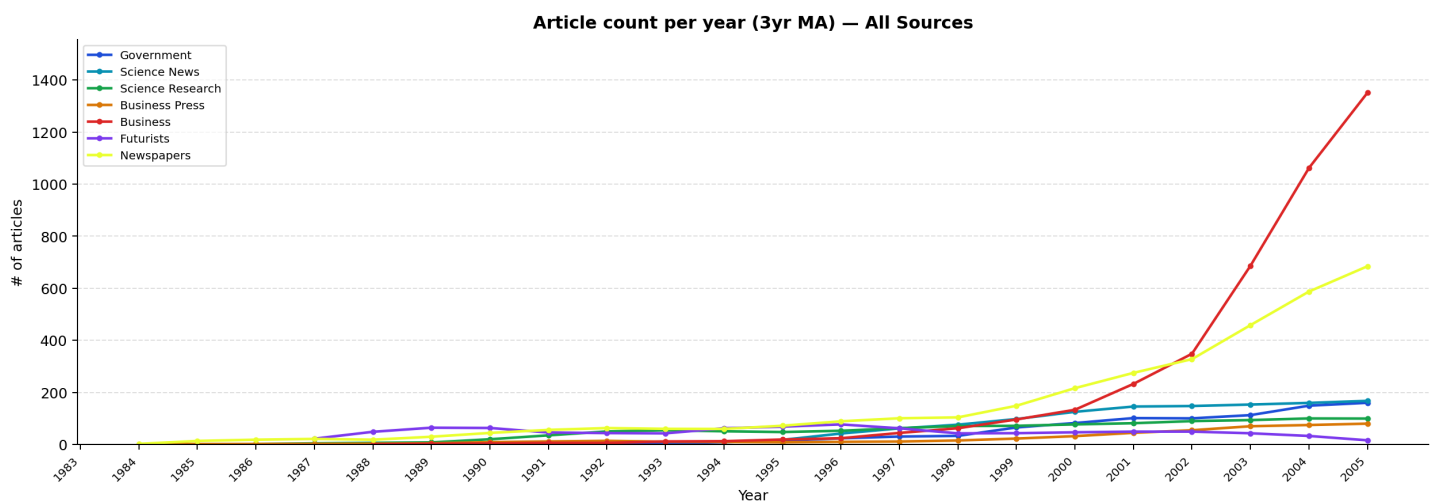
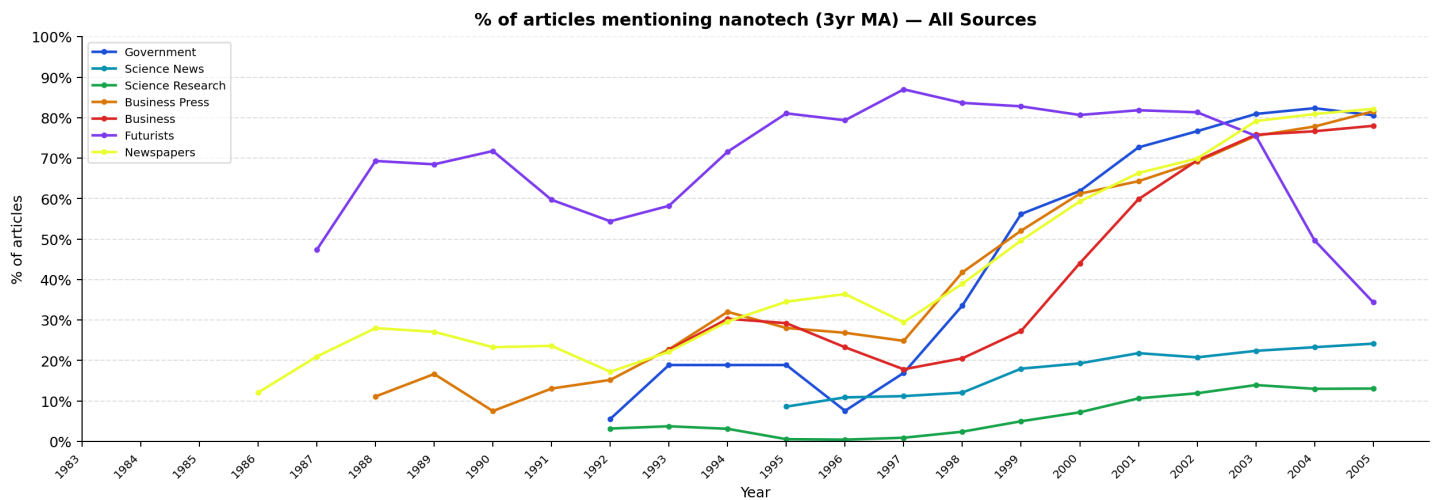
Percentage of coded articles that contain an argument for more nanotechnology funding.



4. Combined all-sources overlay charts

The summary PDF now uses combined all-sources overlay charts instead of per-source individual charts. All 7 sources appear on one plot with 3-year moving averages, making cross-source comparison much easier.

Summary PDF contents (21 figures): 3 combined keyword charts (article count, nanotech %, nanotech total), 1 macro co-occurrence, 4 top-8 co-occurrence (key sources), and 13 qualitative dimension charts.



Next steps

Data gaps:

- Science Research and Newspapers have 0% Claude analysis coverage. Run `python -m qualitative_coding --sheets "Sheet1"` to code all uncoded articles via the API (~\$30-60, ~7,800 articles).
- Science News has only 18% coverage from the old file import.

Quality:

- Calibrate qualitative coding: hand-code 10 articles and compare against Claude's output. Target $\geq 80\%$ agreement on Attitude and Sustaining/Disrupting.

Analysis:

- The qualitative dimension graphs currently cover ~4,300 of 12,200 articles. Running the Claude API on the remaining articles would give full coverage and enable meaningful graphs for all 7 sources.
- Stars vs reference count validation per `next_steps.txt` points 1 and 2 (being done manually).